

PRAGMATIC REWARDS: INTRINSIC COMMUNICATIVE CLARITY IN MULTI-AGENT REINFORCEMENT LEARNING

Henri Langlois, Pablo Piantanida

CNRS, ILLS, Mila, CentraleSupélec

ABSTRACT

Reinforcement Learning (RL) agents often fail in cooperative settings because their actions, while individually optimal, are unpredictable to their partners. This challenge is magnified in human-agent teams, where an agent’s opacity can erode trust and team effectiveness. We address this limitation by establishing a formal equivalence between pragmatic inference, as formalized by the Rational Speech Act (RSA) framework, and Maximum Entropy Reinforcement Learning (Max-Ent RL). This theoretical bridge yields a principled intrinsic reward for “legible” behavior—actions that clearly communicate an agent’s private state—computed offline by a Pragmatic Reward Module (PRM) that adds no latency to decision-making. The framework exposes the literal listener L_0 , the assumed common ground against which clarity is measured, as its central design choice. We study a spectrum of listeners, from hand-crafted geometric heuristics to a listener learned jointly with the policy, and show that the learned instantiation turns the pragmatic reward into a variational lower bound on the mutual information between an agent’s intention and its observable behavior, unifying pragmatic and information-theoretic approaches to implicit communication. Experiments in a cooperative navigation task with private preferences show that hand-crafted listeners induce miscalibrated signaling—costly behavior whose intrinsic reward remains below the uninformative level—degrading task performance by up to 47% while adding no decodable information, whereas the learned listener is reliably optimized at no task cost and exposes a controllable trade-off between information disclosure and task performance. Our framework provides a formal model, and a practical recipe, for agents that are not only effective but also strategically collaborative.

Index Terms— Reinforcement Learning, Intrinsic Motivation, Multi-Agent Systems, Computational Pragmatics, Legibility

1. INTRODUCTION

Human collaboration is fundamentally a goal-oriented, strategic activity. To succeed, we do not just act to achieve our own goals; we act in ways that allow our partners to infer our beliefs and intentions, enabling better coordination. The same is true for artificial agents. In cooperative multi-agent reinforcement learning (MARL) environments, an agent’s success depends not only on its own actions but also on being understood by its teammates. Consider a cooperative navigation task in which each agent must cover a target of a preferred type: an agent that simply rushes toward its target may take a path that is ambiguous from a teammate’s perspective, causing the teammate to abandon a target the agent never intended to claim. A more sophisticated agent might take a slightly suboptimal path that unambiguously singles out its intended target, making its intention clear and enabling better team coordination.

This property of being easily understood is called “legibility” [1]. However, standard RL agents, optimized solely for an extrinsic task reward, have no incentive to be legible. This leads to our central question: how can we formally encourage an RL agent to choose actions that are not just extrinsically optimal for the task, but also intrinsically optimal for communicating its private state or intentions?

This work proposes a solution in the form of an intrinsic reward, R_{comm} , that quantifies the communicative clarity of an agent’s actions. To calculate this reward, we model an “observer” (e.g., another agent) who uses pragmatic inference to reason about why the agent chose a particular action over the available alternatives. We find the ideal formalism for this observer in the Rational Speech Act (RSA) framework [2], which provides a formal account of reasoning about communicative intent.

Any such observer model must be anchored in a literal listener L_0 : the assumed common ground that specifies how behavior would be interpreted in the absence of pragmatic reasoning. Our analysis and experiments show that this choice is not an implementation detail but the decisive factor in whether pragmatic rewards help or hurt. Hand-crafted listeners can be systematically mis-

aligned with what the policy can express, incentivizing costly signaling that an actual observer cannot decode. This motivates our second contribution: replacing the hand-crafted L_0 with a listener learned from the agents’ own behavior, which guarantees, by construction, that rewarded actions are decodable.

Our contributions are as follows:

- We establish a formal equivalence between the objective functions of Max-Ent RL and the RSA framework, allowing the RSA “listener value” to be imported as a principled intrinsic reward for legibility, and we show that purely communicative models such as CRSA [3] are recovered as a myopic special case ($\gamma = 0$, $R_{\text{ext}} = 0$).
- We operationalize the theory as a Pragmatic Reward Module (PRM): an offline component, compatible with centralized training and decentralized execution (CTDE), that computes R_{comm} from replayed transitions and adds no latency at execution time.
- We characterize the space of literal listeners and prove that instantiating L_0 with a listener trained by maximum likelihood on the agents’ behavior makes the expected pragmatic reward a variational lower bound on the mutual information $I(M; S, A)$ between private intentions and observable behavior. This connects pragmatic rewards to information-regularization [4] and skill-discriminability [5] objectives, and positions them as the literal (0-th order) special case of our framework.
- We evaluate the resulting reward family in a cooperative navigation task with private preferences. Hand-crafted listeners exhibit a failure mode we term miscalibrated signaling—agents pay a large task-reward cost for signals that carry no decodable information, detectable as an intrinsic reward stuck below the uninformative level—while the learned listener is reliably optimized at no task cost, and its weight λ traces a controllable information-performance trade-off. Oracle and post-hoc probe comparisons additionally show that the task value of intent transmission can be near zero even when intentions are private, arguing that legibility methods must be evaluated on calibration, information transmitted, and task cost separately.

2. RELATED WORK

Our framework, which uses pragmatic inference to generate an intrinsic reward for communicative clarity, builds on four lines of research: (1) legibility and observer-aware

planning; (2) intrinsic rewards for multi-agent coordination; (3) implicit and emergent communication; and (4) computational pragmatics.

2.1. Legibility, Goal Recognition, and Observer-Aware Planning

Inferring an agent’s latent state (e.g., its goal) from observed behavior is the central problem of Inverse Reinforcement Learning (IRL) [6] and computational Theory of Mind (ToM) [7, 8]. Legibility inverts this problem: the actor shapes its own behavior so that an observer’s inference succeeds. Originally formalized for robot motion [1], legibility has been extended to stochastic sequential decision problems [9], unified with explicability and predictability in observer-aware MDPs [10, 11], and recently brought to MARL through legible variants of interactive POMDPs [12]. These approaches typically embed the observer model in the planning objective, requiring inference over goals at decision time. In contrast, our framework converts legibility into an intrinsic reward computed offline during training, so the executed policy carries its communicative competence “for free,” with no observer model, belief tracking, or additional computation in the execution loop.

A complementary line of work uses ToM as a source of intrinsic motivation [13], rewarding agents for accurately predicting the beliefs of others; this is listener-centric, rewarding agents for being predictive of others. Our pragmatic framework is speaker-centric: it rewards agents for making their own private state clear, regardless of their partners’ modeling ability.

2.2. Intrinsic Rewards for Multi-Agent Coordination

Our pragmatic reward is a form of intrinsic motivation for cooperation, and is thus related to social influence [14], which rewards an agent for causally changing its partners’ behavior. The communicative goals differ: influence rewards an agent for compelling a reaction, whereas the pragmatic reward incentivizes the successful transmission of information about oneself, decoupling the speaker’s reward from the listener’s reaction. Teammates remain free to exploit or ignore the revealed intention, which promotes a more flexible form of coordination.

Closest to our learned-listener instantiation is the information-regularization framework of Strouse et al. [4], which adds a mutual-information term $I(A; G|S)$ to the objective so that agents share (or hide) their goal G through their actions. Similarly, Eccles et al. [15] add a positive-signaling bias encouraging messages to depend informatively on observations, and, in the single-agent setting, DIAYN [5] rewards skills for being discriminable from states via a learned classifier. Proposition 1 makes

the relationship precise: these MI-based objectives correspond to the literal listener level of our framework, obtained when the RSA recursion depth is zero and L_0 is trained by maximum likelihood. The pragmatic levels add contrastive reasoning over the alternative actions the agent could have taken—a model of communication richer than statistical dependence alone. Our framework also shares the goal of grounded, convention-free coordination with Bayesian action decoding [16] and off-belief learning [18], which control how much agents may read into their partners’ actions; we instead directly reward the actor for making the intended reading easy.

2.3. Implicit Communication vs. Explicit Channels

A significant body of MARL research equips agents with a dedicated communication channel and learns an emergent protocol end-to-end [19, 20, 21]. Our framework diverges from this paradigm by focusing on implicit communication: there is no separate channel and no explicit message passing. An agent “speaks” by moving in the environment, and a teammate “listens” by observing that movement. The pragmatic reward incentivizes actions that are simultaneously effective for the task and clear as signals of intent. This is valuable when explicit communication is costly, unavailable, or must be fused with physical action, as is common in human-agent teams. Because the signal is the action, the exploration-communication conflict identified in emergent-communication work [15, 17] appears here as a reward-shaping trade-off, which we analyze through the weight λ .

2.4. Computational Pragmatics

RSA [2] formalizes communication as recursive probabilistic reasoning between a speaker and a listener, and has been given a rate-distortion interpretation [22]. Neural instantiations replace the hand-specified literal listener with learned models [23, 24], a design choice our learned- L_θ instantiation transposes to the behavioral domain. The Collaborative RSA (CRSA) model [3] extends RSA to multi-turn dialog between agents with asymmetric private information. We show in Section 4.6 that such purely communicative models arise as the myopic, task-free special case of our strategic framework, which extends pragmatic reasoning to long-horizon sequential decision-making where instrumental and communicative goals are intertwined.

3. FORMALISMS FOR ACTION AND INTERPRETATION

To model legible strategic behavior we require two formalisms: one for optimal sequential decision-making un-

der uncertainty, and one for how an observer infers intent from an action. For the former we use Max-Ent RL; for the latter, RSA.

3.1. Maximum Entropy RL

Max-Ent RL extends the standard RL objective with the intuition “be as versatile as possible while still being successful.” In a single-step problem, the agent seeks a policy $\pi(a|s)$ maximizing expected reward plus policy entropy [25]:

$$J(\pi) = \mathbb{E}_{a \sim \pi(\cdot|s)}[R(s, a)] + \alpha_{\text{RL}} H(\pi(\cdot|s)). \quad (1)$$

The optimal policy is a softmax over rewards [26], with temperature α_{RL} trading off exploitation against versatility:

$$\pi^*(a|s) \propto \exp(R(s, a)/\alpha_{\text{RL}}). \quad (2)$$

3.2. The Rational Speech Act Framework

RSA [2] models an observer who infers an agent’s intended meaning not from the literal content of an action alone, but by reasoning about why the agent chose that action over the alternatives: “if they had intended X , they would have done Y ; since they did Z , they must intend something else.” Formally, RSA seeks a communicative policy $S(u|m)$ that optimally balances communicative success against action cost, yielding a softmax over the utility, or “listener value,” of each communicative act:

$$S(u|m) \propto \exp(\alpha_{\text{RSA}} \cdot V_L(u, m)), \quad (3)$$

where $V_L(u, m)$ quantifies the clarity of act u for conveying meaning m to an observer, penalized by the act’s cost.

4. REWARDING LEGIBILITY

Our agent optimizes a composite reward balancing task objectives with communicative clarity. This section defines the reward via a formal bridge between pragmatic inference and RL, characterizes the space of listener models that anchor it, describes the architecture that computes it, and presents the resulting strategic Bellman equation.

4.1. The Total Reward

At each timestep t the agent receives the standard extrinsic task reward $R_{\text{ext}}(s_t, a_t)$ from the environment, and generates internally an intrinsic communicative reward $R_{\text{comm}}(s_t, a_t)$ quantifying how clearly action a_t communicates the agent’s private state m_t (its “meaning,” e.g.,

Table 1. Mapping between Max-Ent RL and RSA components.

Max-Ent RL component	RSA analogue
State (s)	Context
Action (a)	Utterance (u)
Private state / intention (m)	Meaning (m)
Agent policy ($\pi(a s, m)$)	Speaker ($S(u m)$)
Intrinsic reward (R_{comm})	Listener value (V_L)
Temperature (α_{RL})	Rationality ($1/\alpha_{\text{RSA}}$)

an intended target) to an imagined observer. The policy maximizes the weighted sum

$$R_{\text{total}}(s_t, a_t) = R_{\text{ext}}(s_t, a_t) + \lambda R_{\text{comm}}(s_t, a_t), \quad (4)$$

where λ balances task success against communicative clarity. The central question is how to define R_{comm} in a principled way.

4.2. Deriving R_{comm} via Pragmatic Inference

We treat the agent’s action a as a communicative “utterance” and its private state m as the “meaning” to be conveyed (Table 1). At first glance, the objective of an RL agent and the goal of a pragmatic speaker seem unrelated. However, the optimal single-step Max-Ent policy (Eq. 2) and the RSA speaker (Eq. 3) are both softmax distributions, identical under the re-indexing $\alpha_{\text{RSA}} = 1/\alpha_{\text{RL}}$. This formal alignment is the cornerstone of our framework: it justifies importing the well-defined listener value V_L from computational pragmatics as a principled intrinsic reward. We define

$$R_{\text{comm}}(s_t, a_t) \triangleq V_L(a_t, m_t) = \log L_t^*(m_t|a_t, s_t) - C(a_t), \quad (5)$$

where $L_t^*(m_t|a_t, s_t)$ is the probability a pragmatic listener assigns to the agent’s true private state after observing its action in context s_t , and $C(a_t)$ is an optional action cost. In practice we set $C(a_t) = 0$: the cost of a legible action is already handled by the task reward, since a less direct but clearer path is naturally penalized by a lower R_{ext} . In our implementation we also add the constant $\log |\mathcal{M}|$, so that R_{comm} reads as the information gain over a uniform prior: it is zero when the action is uninformative and positive when the listener’s posterior concentrates on the true meaning.

4.3. The Space of Literal Listeners

Computing L^* requires anchoring the pragmatic recursion in a literal listener $L_0(m|a, s)$: the assumed common ground specifying how behavior would be interpreted absent pragmatic reasoning. Given L_0 and a set $\mathcal{A}$ of al-

ternative actions (the true action plus K sampled alternatives), the pragmatic listener is obtained by N_r steps of RSA alternation,

$$S_k(a|m) = \frac{L_{k-1}(m|a)}{\sum_{a' \in \mathcal{A}} L_{k-1}(m|a')}, \quad L_k(m|a) = \frac{S_k(a|m)}{\sum_{m'} S_k(a|m')}, \quad (6)$$

and $L^* = L_{N_r}$. The choice of L_0 is the framework’s most consequential free parameter, and we study three instantiations.

(i) Hand-crafted geometric listener. A computationally cheap heuristic interprets actions by their geometric similarity to ideal paths. For agent position p and each meaning m , let v_m be the unit vector toward the nearest target consistent with m . A naïve score is $S(a, m) = \alpha_{\text{temp}} \cos(a, v_m)$. To capture unambiguity explicitly, an exclusivity variant penalizes similarity to competing meanings,

$$S_{\text{excl}}(a, m) = \alpha_{\text{temp}} (\cos(a, v_m) - \max_{m' \neq m} \cos(a, v_{m'})), \quad (7)$$

and $L_0(\cdot|a) = \text{softmax}_m S_{\text{excl}}(a, \cdot)$. This listener rewards actions aligned with the ideal path toward the true target and misaligned with the ideal paths of all confusable targets.

(ii) Learned listener. Hand-crafted listeners presuppose that the features they score are the ones an actual observer would use. To remove this assumption, we train a network $L_\theta(m|s, a)$ by maximum likelihood on the replay buffer to predict an agent’s true meaning from its publicly observable context and action (the private state is masked from the input). The listener is trained concurrently with the policy, so the common ground co-adapts: the reward only credits signals that are actually decodable from the agents’ current behavioral distribution. With $N_r = 0$ the reward is simply $R_{\text{comm}} = \log L_\theta(m_t|s_t, a_t) + \log |\mathcal{M}|$; with $N_r \geq 1$ the RSA recursion (Eq. 6) is applied on top of L_θ , adding contrastive reasoning over alternatives.

The learned instantiation admits an information-theoretic characterization.

Proposition 1. Let meanings be drawn uniformly on $\mathcal{M}$, and let $L_\theta(m|s, a)$ be any conditional distribution. Then, under the joint distribution induced by the policy,

$$\mathbb{E}[\log L_\theta(m|s_t, a_t) + \log |\mathcal{M}|] \leq I(M; S_t, A_t), \quad (8)$$

with equality iff $L_\theta(m|s, a) = p(m|s_t = s, a_t = a)$. Hence the expected learned-listener reward is the Barber–Agakov variational lower bound [27] on the mutual information between the agent’s private meaning and its observable behavior; maximum-likelihood training of θ tightens the bound, and policy optimization against it maximizes information disclosure.

Proof. $\mathbb{E}[\log L_\theta(m|s, a)] = \mathbb{E}[\log p(m|s, a)] - \mathbb{E}_{s,a}[\text{KL}(p(\cdot|s, a) \| L_\theta(\cdot|s, a))]$. Adding $H(M) = \log |\mathcal{M}|$ gives the claim; the KL term vanishes iff L_θ matches the true posterior. $\square$

Proposition 1 unifies two families of methods. The cooperative information regularizer of Strouse et al. [4], the positive-signaling bias of Eccles et al. [15], and the skill discriminator of DIAYN [5] all maximize variational MI bounds of exactly this form; within our framework they are the literal ($N_r = 0$) level of pragmatic reasoning with a learned common ground. Conversely, the RSA recursion is a strict generalization that scores an action not by how much information it carries in isolation, but by how it contrasts with the alternatives the agent declined—the signature of pragmatic communication [2, 22].

4.4. The Pragmatic Reward Module

The theoretical bridge yields a practical hybrid architecture that decouples action selection from communicative analysis. The system comprises the main RL agent, which learns $\pi(a_t|s_t)$, and a Pragmatic Reward Module (PRM), which computes R_{comm} to shape the agent’s learning updates. The pragmatic analysis is performed offline during training—for static listeners, when a transition is collected; for the learned listener, when it is replayed, so the reward tracks the co-adapting common ground—and never in the action-selection loop, adding no latency to decision-making. We operate under CTDE: during centralized training the PRM has access to all agents’ private states, enabling a gold-standard legibility reward; at execution, agents act from local observations only, and the PRM is discarded.

4.5. The Strategic Bellman Equation

With the composite reward defined, the action-value function of the legible agent satisfies the soft Bellman equation

$$Q_t(s_t, a_t) = R_{\text{total}}(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1}} \left[\alpha \log \sum_{a_{t+1}} \exp \left(\frac{1}{\alpha} Q_{t+1}(s_{t+1}, a_{t+1}) \right) \right], \quad (9)$$

which is the standard Max-Ent backup applied to R_{total} . The equation makes explicit the trade-off between immediate gains (task and clarity) and the long-term strategic value of future states: an agent may accept a lower immediate reward (e.g., a “corralling” detour) if the resulting state carries higher expected value through improved team coordination.

Purely communicative agents arise as a special case. Setting $R_{\text{ext}} = 0$ and $\lambda = 1$ makes the total reward purely communicative; setting $\gamma = 0$ makes the agent myopic, collapsing Eq. 9 to the immediate reward plus policy entropy. This objective is exactly the per-turn gain function of the Collaborative RSA model [3], demonstrating that single-turn pragmatic models are a specific instance of our framework, which extends pragmatic reasoning to long-horizon sequential settings where instrumental and communicative goals interact.

5. EXPERIMENTAL SETUP

5.1. Environment

We evaluate on a cooperative navigation task with private preferences, a modified Simple Spread scenario in the spirit of the Multi-Agent Particle Environments [28]. Three agents and three landmarks occupy a continuous 2D arena; each landmark carries a color (a random permutation of three categories per episode) and each agent holds a private preference m_i for one color, drawn uniformly at random. The team reward at each step comprises (a) a preference bonus for every landmark covered by an agent whose preference matches the landmark’s color, (b) a distance penalty equal to the sum, over landmarks, of the distance to the nearest agent, and (c) a collision penalty per overlapping pair. Actions are continuous 2D forces; episodes last 50 steps. Each agent observes its own position and velocity, its preference (one-hot), landmark relative positions and colors, and teammates’ relative positions and velocities—but not teammates’ preferences. Coordination therefore hinges on inferring who is claiming what from motion alone: an agent must both act on its own preference and legibly signal it, so that teammates can confidently cover the remaining landmarks.

5.2. Models and Conditions

All agents are trained with multi-agent Soft Actor-Critic (MASAC)—parameter-shared Gaussian actors and a centralized twin critic over joint observations and actions, with automatic entropy tuning—the natural multi-agent extension of the Max-Ent framework our theory is built on. We compare six conditions that differ only in the reward and observation content:

- Baseline: $\lambda = 0$; extrinsic reward only.
- Oracle: $\lambda = 0$, but observations include teammates’ true preferences. Since no inference is needed, the oracle-baseline gap measures the task value of intent information itself.

- PRM (exclusivity L_0): the hand-crafted geometric listener with exclusivity scoring (Eq. 7) and one step of RSA recursion over $K = 64$ sampled alternative actions.
- PRM (simple L_0): identical but without the exclusivity term, i.e., plain cosine scoring.
- PRM (learned): the learned listener L_θ at recursion depth $N_r = 0$; L_θ is a 2-layer MLP trained by cross-entropy on replayed transitions, with the private preference masked from its input.
- PRM (learned + RSA): the learned listener with one step of RSA recursion over 16 sampled alternatives.

All PRM conditions use $\lambda = 0.1$ applied to the per-team sum of agents’ communicative rewards. Each condition is run with 3 random seeds for 480k environment steps (64 vectorized environments). Full hyperparameters are listed in Table 3.

5.3. Evaluation Metrics

We evaluate periodically on 64 held-out episodes with deterministic policies. Task performance is the mean per-step team reward and its components. Team specialization is the fraction of landmarks claimed uniquely (each agent assigned to its nearest landmark); a score of 1 indicates perfect division of labor. Legibility is measured by a fixed probe listener—the simple geometric listener, used identically across all conditions—reporting the accuracy with which an agent’s true preference is recovered from its instantaneous action (probe intent accuracy). The probe is computed over all steps, including those where an agent already rests on its target and its actions are uninformative, so absolute values understate legibility; because the probe is fixed and external to all training conditions, however, relative improvements reflect genuinely clearer behavior rather than adaptation to the metric. For learned-listener conditions we additionally report L_θ ’s own held-out accuracy, and for all PRM conditions the achieved communicative reward R_{comm} . Finally, we measure post-hoc decodability: for every trained run, a fresh listener network is trained from scratch on deterministic rollouts of the final policy and evaluated on held-out episodes. This quantifies how much intent information each condition’s converged behavior carries under an identical decoding budget, independently of any listener used during training.

6. RESULTS

Table 2 summarizes final performance; Figures 3–6 show training dynamics. The results separate cleanly along

Table 2. Final performance (mean $\pm$ s.e.m. over 3 seeds, averaged over the last three evaluations). R_{ext} is the per-step team reward; Spec. is team specialization; R_{comm} is each condition’s own communicative reward (information-gain scale); Post-hoc is the decoding accuracy of a fresh listener trained on the final policy’s rollouts (chance = .33).

Condition	R_{ext}	Spec.	R_{comm}	Post-hoc
Baseline	$1.07 \pm .03$	.847	—	$.598 \pm .02$
Oracle	$0.99 \pm .04$	.841	—	$.589 \pm .01$
Exclusivity L_0	$0.57 \pm .04$	.815	−2.39	$.604 \pm .02$
Simple L_0	$0.93 \pm .06$	.833	−1.33	$.599 \pm .01$
Learned L_θ	$1.03 \pm .08$	.847	−0.13	$.582 \pm .03$
Learned + RSA	$1.09 \pm .03$	.853	−0.13	$.578 \pm .02$

the common-ground axis: what matters is not whether a pragmatic reward is used, but which listener anchors it.

6.1. Hand-Crafted Listeners: Miscalibrated Signaling

The exclusivity listener exhibits an instructive failure mode, which we also observed in preliminary experiments with an independent, larger-scale implementation. The composite reward is clearly operative—early in training it even accelerates learning (Fig. 3), and it drives the geometric probe accuracy up faster than any other condition (Fig. 5)—yet the agent never comes close to optimizing the communicative reward itself. R_{comm} plateaus around −2.4 (Fig. 6), below the uninformative level of zero: the pragmatic listener built on this common ground assigns the true meaning less probability than a uniform guess. The exclusivity score (Eq. 7) demands that an action align with the true target’s direction while misaligning with every competing direction; when landmarks subtend similar bearings this margin is geometrically unsatisfiable, so the agent pays a permanent tax for a signal that cannot be produced. The tax is visible everywhere: final task reward drops by 47% relative to the baseline, distance and collision penalties rise, and specialization saturates lower (.815 vs. .847). The simple cosine listener, which removes the exclusivity margin, is correspondingly less harmful ($R_{\text{ext}} = 0.93$) and achieves a better—but still negative— R_{comm} . We call this failure mode miscalibrated signaling: a hand-crafted common ground rewards behavior the policy cannot realize and an observer cannot decode, converting the intrinsic reward into pure interference.

6.2. Learned Listeners: Calibrated Legibility

The learned listener changes the picture qualitatively. Both learned conditions steadily optimize their commu-

nicative reward from -0.65 toward zero (Fig. 6), confirming that the co-adapted reward is learnable—by construction, it only credits distinctions that are actually present in the behavioral distribution. This calibration eliminates the destructive interference: task performance is statistically indistinguishable from the baseline ($1.03 \pm .08$ and $1.09 \pm .03$ vs. $1.07 \pm .03$), with the pragmatic variant nominally best on task reward and specialization. During training, the co-adapting listener’s accuracy climbs from chance ($1/3$) to $\approx .43$ and is still rising at the end (Fig. ??), with the RSA-recursed variant consistently ahead of the purely literal one—suggesting that contrastive reasoning over alternatives extracts signal that maximum-likelihood decoding alone does not.

6.3. How Much Was There to Say? Oracles and Post-Hoc Probes

Two further measurements put the above in perspective. First, the oracle: giving agents their teammates’ true preferences does not improve task performance (0.99 vs. 1.07). Near-optimal coordination in this environment is achievable from positions and velocities alone—who is closest to what resolves most conflicts—so the marginal task value of intent information is small. Second, the post-hoc probes: a fresh listener trained on any condition’s converged behavior decodes intentions at $.58$ – $.60$ accuracy (Table 2), with no significant differences across conditions. Because acting on one’s preference is extrinsically rewarded, converged behavior is already informative; at $\lambda = 0.1$ there is simply little decodable information left for a calibrated reward to add—which the learned listener correctly reflects by saturating near zero, and which the hand-crafted listeners ignore by charging a permanent task tax for no informational return.

Together these results reframe the evaluation of implicit-communication methods: task reward alone conflates the cost of signaling with its value, and a method should be judged on (a) whether its intrinsic reward is achievable (calibration), (b) how much decodable information the resulting behavior carries (post-hoc probes), and (c) what it pays for that information (task-reward gap). On all three criteria the learned listener dominates the hand-crafted ones.

6.4. The Weight λ : Dial or Landmine?

We sweep the communicative weight for both listener families (2 seeds per new point; Fig. 1). With the learned listener, λ behaves like a dial along an information-performance frontier: $\lambda = 0.1$ is free ($R_{\text{ext}} = 1.03$, $R_{\text{comm}} = -0.13$); $\lambda = 0.3$ buys near-saturated clarity ($R_{\text{comm}} = -0.05$) for a modest task cost ($R_{\text{ext}} = 0.84$); and $\lambda = 1.0$ pushes the communicative reward to genuinely positive information gain ($R_{\text{comm}} = +0.62$)—

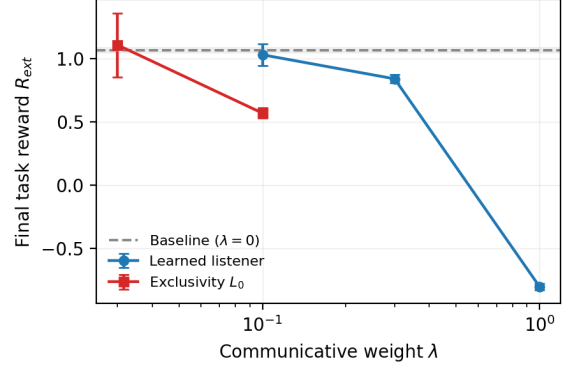


Fig. 1. Final task reward vs. communicative weight λ . The learned listener trades task reward for information smoothly; the exclusivity listener offers no λ at which information is gained.

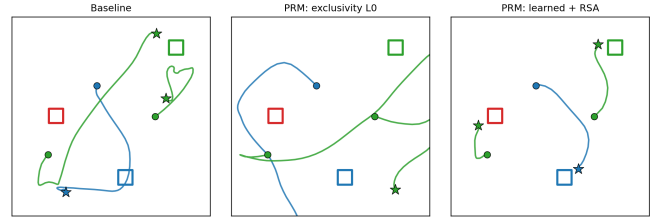


Fig. 2. A held-out episode under three conditions. Squares are landmarks (color = category); curves are agent trajectories colored by the agent’s private preference; circles mark start positions, stars final positions.

behavior more informative than task-optimal motion—at a large task cost ($R_{\text{ext}} = -0.80$) as agents privilege signaling over covering landmarks. With the exclusivity listener, no operating point buys information: at $\lambda = 0.03$ the reward is simply ignored (R_{comm} stuck at -2.6 , task performance at baseline level), and at $\lambda = 0.1$ it destroys 47% of task reward with R_{comm} still at -2.4 . Calibration is precisely what turns λ from a landmine into a dial.

6.5. Qualitative Behavior

Figure 2 shows a representative held-out episode. The baseline reaches landmarks but meanders, with late target switches. The exclusivity-listener agent displays the miscalibration diagnosed above: trajectories sweep across the arena and settle far from any landmark—costly motion signaling nothing. The learned-pragmatic agent produces direct, mutually distinct paths: each agent claims a separate landmark, and the agent whose preferred landmark is taken by a teammate cleanly re-targets the remaining one.

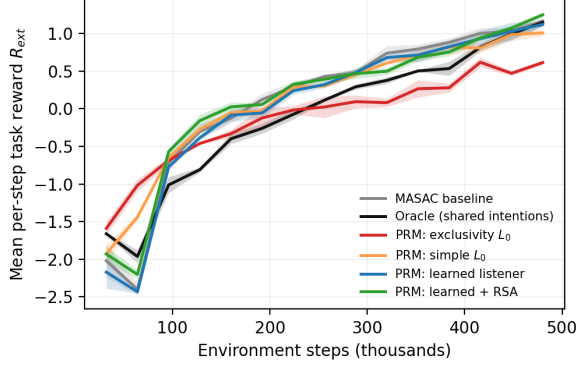


Fig. 3. Mean per-step team reward R_{ext} (3 seeds, mean $\pm$ s.e.m.).

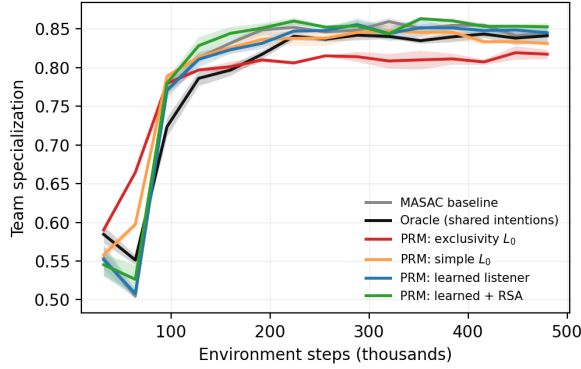


Fig. 4. Team specialization: fraction of landmarks claimed uniquely.

7. DISCUSSION

The common ground is the problem. Our central empirical finding is that the success of a pragmatic reward hinges almost entirely on the alignment between the literal listener and the agents’ realizable behavior. A hand-crafted L_0 encodes the designer’s theory of what signaling should look like; when that theory is even mildly wrong—as with the exclusivity margin, which demands geometrically unsatisfiable action patterns—the recursion built on top of it faithfully amplifies the error, and the agent is paid to produce elaborate, undecodable gestures. This failure is quiet: behavioral metrics (path deviation, action magnitude, early probe accuracy) suggest the agent is “signaling harder,” while the communicative channel carries nothing. We suspect this failure mode is widespread in reward designs that score behavior against idealized observer models, and recommend a simple diagnostic that requires no extra machinery: track whether the agent can drive its own intrinsic reward above the uninformative level. A persistently negative information

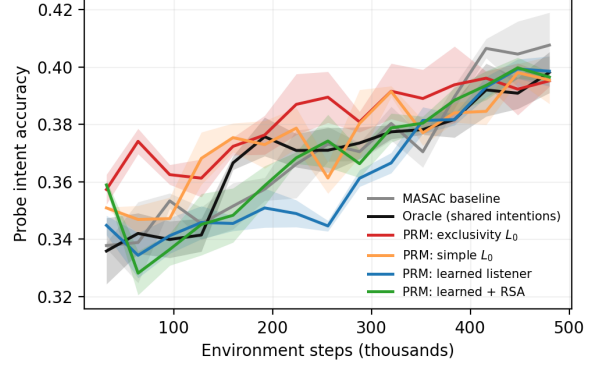


Fig. 5. Probe intent accuracy under the fixed geometric probe listener, identical across conditions.

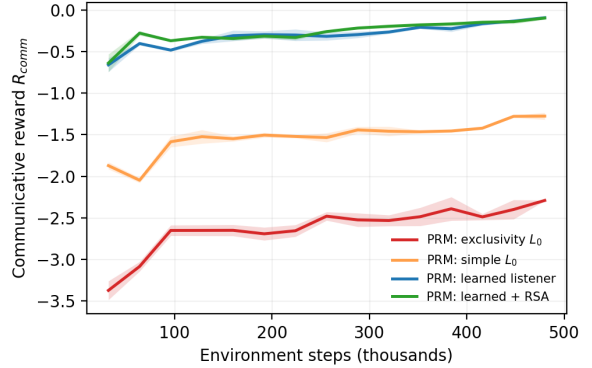


Fig. 6. Achieved communicative reward R_{comm} (each condition under its own listener; information-gain scale, 0 = uninformative).

gain is a certificate of miscalibration.

Learning the listener closes the loop. Training L_θ on the agents’ own replay data guarantees that the common ground never demands what the policy distribution does not contain; the reward then only sharpens distinctions that already exist, which is why it optimizes smoothly and costs no task performance. Proposition 1 explains the mechanism: the learned-listener reward is a variational mutual-information bound, so policy improvement against it is information maximization—grounded, by construction, in a decodable channel. And because the bound is calibrated, its weight becomes meaningful: λ moves the policy along an actual information-performance frontier (Sec. 6.4) instead of scaling an arbitrary shaping term. The RSA recursion contributes during training: it consistently improves the co-adapting listener’s accuracy, suggesting that pragmatic contrast (scoring an action against foregone alternatives) accelerates the emergence of decodable behavior, even where

asymptotic decodability is capped by the task.

When does legibility pay? The oracle result is a caution for the field: in environments where geometry resolves coordination, intent transmission has little task value to unlock, and a well-calibrated legibility reward should—and in our experiments does—approach a no-op on task reward while still making behavior more decodable. The task-level benefits of legibility should be sought where intentions are both hidden and consequential: asymmetric capabilities, longer horizons where early commitment matters, or human partners who cannot run an optimal filter over observed motion. Designing such benchmarks, in which the oracle-baseline gap is provably large, is in our view a prerequisite for meaningful progress on communicative MARL.

7.1. Limitations and Future Work

Several limitations qualify our findings and chart future work. First, the intrinsic reward densifies the learning signal, which can aid learning independently of legibility; our probe-based metrics mitigate but do not eliminate this confound. Second, meanings here are hand-specified discrete variables; extending R_{comm} to continuous, hierarchical, or self-assigned meaning spaces—where the agent learns what is worth communicating, not only how—remains open. Third, our validation is confined to AI-AI teams in 2D particle environments; whether computational legibility transfers to human-perceived legibility must be established with human observers, for instance by fitting L_θ to human intent judgments, thereby aligning the intrinsic reward with what people actually find legible. Fourth, CTDE grants the PRM privileged access to private states during training; fully decentralized variants would need agents to estimate each other’s meanings. Finally, the framework is symmetric in an intriguing way: inverting the sign of λ turns legibility into obfuscation, suggesting applications to privacy-preserving and adversarial settings where agents should remain opaque to opponents while staying legible to allies [4].

8. CONCLUSION

We presented a framework that unifies pragmatic inference and maximum-entropy reinforcement learning, importing the RSA listener value as a principled intrinsic reward for legible behavior. The framework isolates the literal listener—the assumed common ground—as the decisive design choice, and our analysis shows that learning this listener jointly with the policy both aligns the reward with what observers can actually decode and recovers information-theoretic approaches to implicit communication as a special case. Empirically, hand-crafted lis-

teners induce costly but undecodable signaling, whereas the learned listener produces calibrated legibility. We view this as a step toward agents whose competence includes being understood—a prerequisite for fluent coordination with artificial and human partners alike.

A. HYPERPARAMETERS

Table 3. Hyperparameters (all conditions unless noted).

Parameter	Value
Parallel environments	64
Episode length	50
Total environment steps	480k
Actor / critic architecture	MLP 256×256
Learning rate (actor, critic)	3×10^{-4}
Discount γ	0.95
Target smoothing τ	0.005
Batch size / updates per env step	512 / 4
Entropy target	$-\dim(\mathcal{A})$ (auto-tuned)
λ (pragmatic weight)	0.1
Listener L_θ	MLP 128×128 , lr 10^{-3}
Heuristic listener temp. $\alpha_{\text{temp}} / K / N_r$	5 / 64 / 1
Learned+RSA alternatives / N_r	16 / 1
Seeds per condition	3

B. REFERENCES

- [1] Anca D. Dragan, Kenton C.T. Lee, and Siddhartha S. Srinivasa, “Legibility and predictability of robot motion,” in ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2013, pp. 301–308.
- [2] Noah D. Goodman and Michael C. Frank, “Pragmatic language interpretation as probabilistic inference,” *Trends in Cognitive Sciences*, vol. 20, no. 11, pp. 818–829, 2016.
- [3] Lautaro Estienne, Gabriel Ben Zenou, Nona Naderi, Jackie Chi Kit Cheung, and Pablo Piantanida, “Collaborative rational speech act: Pragmatic reasoning for multi-turn dialog,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
- [4] DJ Strouse, Max Kleiman-Weiner, Josh Tenenbaum, Matt Botvinick, and David J. Schwab, “Learning to share and hide intentions using information regularization,” in *Advances in Neural Information Processing Systems*, 2018, vol. 31.
- [5] Benjamin Eysenbach, Abhishek Gupta, Julian Ibarz, and Sergey Levine, “Diversity is all you need: Learning skills without a reward function,” in *International Conference on Learning Representations*, 2019.
- [6] Brian D. Ziebart, Andrew L. Maas, J. Andrew Bagnell, and Anind K. Dey, “Maximum entropy inverse reinforcement learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2008, pp. 1433–1438.
- [7] Chris L. Baker, Julian Jara-Ettinger, Rebecca Saxe, and Joshua B. Tenenbaum, “Rational quantitative attribution of beliefs, desires and percepts in human mentalizing,” *Nature Human Behaviour*, vol. 1, no. 4, pp. 0064, 2017.
- [8] Neil C. Rabinowitz, Frank Perbet, H. Francis Song, Chiyuan Zhang, S.M. Ali Eslami, and Matthew Botvinick, “Machine theory of mind,” in *International Conference on Machine Learning*, 2018, pp. 4218–4227.
- [9] Miguel Faria, Francisco S. Melo, and Ana Paiva, ““Guess what I’m doing”: Extending legibility to sequential decision tasks,” *Artificial Intelligence*, vol. 330, pp. 104107, 2024.
- [10] Shuwa Miura and Shlomo Zilberstein, “A unifying framework for observer-aware planning and its complexity,” in *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2021, pp. 610–620.
- [11] Shuwa Miura and Shlomo Zilberstein, “Observer-aware planning with implicit and explicit communication,” in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2024, pp. 1409–1417.
- [12] Yanyu Liu, Yinghui Pan, Yifeng Zeng, Biyang Ma, and Prashant Doshi, “Active legibility in multiagent reinforcement learning,” *Artificial Intelligence*, vol. 346, pp. 104357, 2025.
- [13] Ini Oguntola, Joseph Campbell, Simon Stepputtis, and Katia Sycara, “Theory of mind as intrinsic motivation for multi-agent reinforcement learning,” in *ICML Workshop on Theory of Mind in Communicating Agents*, 2023.
- [14] Natasha Jaques, Angeliki Lazaridou, Edward Hughes, Caglar Gulcehre, Pedro A. Ortega, DJ Strouse, Joel Z. Leibo, and Nando de Freitas, “Social influence as intrinsic motivation for multi-agent deep reinforcement learning,” in *International Conference on Machine Learning*, 2019, pp. 3040–3049.
- [15] Tom Eccles, Yoram Bachrach, Guy Lever, Angeliki Lazaridou, and Thore Graepel, “Biases for emergent communication in multi-agent reinforcement learning,” in *Advances in Neural Information Processing Systems*, 2019, vol. 32.

- [16] Jakob N. Foerster, Francis Song, Edward Hughes, Neil Burch, Iain Dunning, Shimon Whiteson, Matthew Botvinick, and Michael Bowling, “Bayesian action decoder for deep multi-agent reinforcement learning,” in International Conference on Machine Learning, 2019, pp. 1942–1951.
- [17] Hengyuan Hu and Jakob N. Foerster, “Simplified action decoder for deep multi-agent reinforcement learning,” in International Conference on Learning Representations, 2020.
- [18] Hengyuan Hu, Adam Lerer, Brandon Cui, Luis Pineda, Noam Brown, and Jakob Foerster, “Off-belief learning,” in International Conference on Machine Learning, 2021, pp. 4369–4379.
- [19] Jakob Foerster, Ioannis Alexandros Assael, Nando de Freitas, and Shimon Whiteson, “Learning to communicate with deep multi-agent reinforcement learning,” in Advances in Neural Information Processing Systems, 2016, vol. 29.
- [20] Sainbayar Sukhbaatar, Arthur Szlam, and Rob Fergus, “Learning multiagent communication with backpropagation,” in Advances in Neural Information Processing Systems, 2016, vol. 29.
- [21] Brendon Boldt and David Mortensen, “A review of the applications of deep learning-based emergent communication,” *Transactions on Machine Learning Research*, 2024.
- [22] Noga Zaslavsky, Jennifer Hu, and Roger P. Levy, “A rate-distortion view of human pragmatic reasoning,” in *Proceedings of the Society for Computation in Linguistics*, 2021, pp. 347–348.
- [23] Jacob Andreas and Dan Klein, “Reasoning about pragmatics with neural listeners and speakers,” in *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2016, pp. 1173–1182.
- [24] Daniel Fried, Jacob Andreas, and Dan Klein, “Unified pragmatic models for generating and following instructions,” in *Proceedings of NAACL-HLT*, 2018, pp. 1951–1963.
- [25] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” in International Conference on Machine Learning, 2018, pp. 1861–1870.
- [26] Sergey Levine, “Reinforcement learning and control as probabilistic inference: Tutorial and review,” *arXiv preprint arXiv:1805.00909*, 2018.
- [27] David Barber and Felix Agakov, “The IM algorithm: A variational approach to information maximization,” in *Advances in Neural Information Processing Systems*, 2003, vol. 16.
- [28] Ryan Lowe, Yi Wu, Aviv Tamar, Jean Harb, Pieter Abbeel, and Igor Mordatch, “Multi-agent actor-critic for mixed cooperative-competitive environments,” in *Advances in Neural Information Processing Systems*, 2017, vol. 30.